PANJAB UNIVERSITY, CHANDIGARH

(Estted. under the Panjab University Act VII of 1947- enacted by the Govt. of India)

FACULTY OF ARTS

SYLLABI

FOR

MASTERS IN REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS (RS & GIS) (SEMESTER SYSTEM) EXAMINATIONS, 2015-16

GUIDELINES FOR CONTINUOUS INTERNAL ASSESSMENT (20%) FOR REGULAR STUDENTS OF MASTERS IN REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS (RS & GIS)
(Semester System)
(Effective from the First Year Admissions for the Academic Session 2014-2015)

1. The Syndicate has approved the following guidelines, mode of testing and evaluation including Continuous Internal Assessment of students :

   (i) Terminal Evaluation :80 % (Theory 50% and Practical 30% as specified in syllabus)

   (ii) Continuous Assessment : 20 %

   (iii) Continuous Assessment may include written assignment, snap tests, participation in discussions in the class, term papers, attendance etc.

   (iv) In order to incorporate an element of Continuous Internal Assessment of students, the Colleges/Departments will conduct the following:

(a)	Written Test	:	25 (reduced to 5)
(b)	Snap Test	:	25 (reduced to 5)
(c)	Participation in Class discussion	:	15 (reduced to 3)
(d)	Term Paper	:	25 (reduced to 5)
(e)	Attendance	:	10 (reduced to 02)

Total: 100 reduced to 20

2. Weightage of 2 marks for attendance component out of 20 marks for Continuous Assessment shall be available only to those students who attend 75% and more of classroom lectures/seminars/workshops. The break–up of marks for **attendance component** for theory papers shall be as under :

Attendance Component			Mark/s for Theory Papers
(a)	75 % and above upto 85 %	:	1
(b)	Above 85 %	:	2

3. It shall **not be compulsory** to pass in Continuous Internal Assessment. Thus, whatever marks are secured by a student out of 20 marks, will be carried forward and added to his/her score out of 80, i.e. the remaining marks allocated to the particular subject and, thus, he/she shall have to secure pass marks both in the University examinations as well as total of Internal Continuous Assessment and University examinations.

4. Continuous Internal Assessment awards from the affiliated Colleges/Departments must be sent to the Controller of Examinations, by name, **two weeks before** the commencement of the particular examination on the *proforma* obtainable from the Examination Branch.

PANJAB UNIVERSITY, CHANDIGARH

Outlines of tests, syllabi and courses of reading for Masters in Remote Sensing and Geographic Information Systems (RS & GIS) (Semester System) for the examinations of 2015-16.

SEMESTER-I			
Paper	**Title**	**Max. Marks**	**Credits**
I	Fundamentals of Remote Sensing (a) Theory : 50 (b) Practical (Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30 (c) Internal Assessment : 20	100	4
II	Fundamentals of Photogrammetry (a) Theory : 50 (b) Practical (Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30 (c) Internal Assessment : 20	100	4
III	Fundamentals of Geographic Information Systems (GIS) (a) Theory : 50 (b) Practical (Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30 (c) Internal Assessment : 20	100	4
IV	Fundamentals of Computers and Computer Programming (a) Theory : 50 (b) Practical (Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30 (c) Internal Assessment : 20	100	4
	Total	**400**	**16**
SEMESTER-II			
Paper	**Title**	**Max. Marks**	**Credits**
V	Cartography (a) Theory : 50 (b) Practical (Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30 (c) Internal Assessment : 20	100	4
VI	Spatial Analysis (a) Theory : 50 (b) Practical (Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30 (c) Internal Assessment : 20	100	4
VII	Image Processing (a) Theory : 50 (b) Practical (Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30 (c) Internal Assessment : 20	100	4
VIII	Fundamentals of Global Positioning System (GPS) (a) Theory : 50 (b) Practical (Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30 (c) Internal Assessment : 20	100	4
	Total	**400**	**16**

SEMESTER-III			
Paper	**Title**	**Max. Marks**	**Credits**
IX	Remote Sensing and Geographic Information Systems in Geosciences (a) Theory : 50 (b) Practical (Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30 (c) Internal Assessment : 20	100	4
X	Remote Sensing and Geographic Information Systems in Urban and Regional Planning (a) Theory : 50 (b) Practical (Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30 (c) Internal Assessment : 20	100	4
XI	Research Methodology (a) Theory : 80 (b) Internal Assessment : 20	100	4
XII	Internship and Field Report (a) Evaluation of the Internship and Field Report : 50 (b) Presentation of Data Analysis and Main Findings : 30 (c) Viva-voce : 20	100	4
	Total	**400**	**16**
SEMESTER-IV			
Paper	**Title**	**Max. Marks**	**Credits**
XIII	Project Report 1. Formulation of Project Proposal 2. Mid Term appraisal 3. Pre-submission Presentation 4. Evaluation 5. Presentation 6. Viva-voce	 50 50 50 150 50 50	
	Total	**400**	**16**
	GRAND TOTAL (Semester-I, II, III, IV)	**1600**	**64**

NOTE:

1. Those candidates, who wish to discontinue after successful completion of first two semesters of the Masters course shall be awarded Diploma in Remote Sensing and Geographic Information Systems (RS & GIS).
2. The minimum marks required to pass the examination shall be 45% in each (i) theory paper, (ii) practical work, and (iii) Internship and field/project report, and 50% in aggregate.
3. The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

SEMESTER - I

PAPER-I: FUNDAMENTALS OF REMOTE SENSING

Max. Marks	**: 100**
Theory	: 50
Practical	: 30
Internal Assessment	: 20

OBJECTIVES: To introduce the basic principles of remote sensing, satellite systems and their functioning, and applications of remote sensing technology.

(A) THEORY **Marks: 50**
Time: 3 Hours

UNIT – I

Fundamental Concepts of Remote Sensing:

i. Remote Sensing: Definition, Concept, History and Applications
ii. Types of Remote Sensing
iii. Remote Sensing Platforms and Scanning Systems

UNIT – II

EMR Principles and Interaction Mechanisms:

iv. Radiation Principles; Electromagnetic Spectrum
v. Energy-Atmosphere Interaction; Atmospheric Windows
vi. Energy-Earth Interaction; Spectral Signatures of Surface Features

UNIT – III

Image Processing and Interpretation:

vii. Image: Meaning and Types (Analogue and Digital) and Characteristics
viii. Resolution: Spatial, Spectral, Radiometric and Temporal
ix. Basics of Image Processing; Elements of Image Interpretation.

UNIT – IV

Satellite Systems and Sensors:

x. LANDSAT, SPOT and IRS
xi. IKONOS, Quickbird, World-View and GeoEye
xii. SRTM, ASTER, MODIS

NOTE:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Comparing Satellite Imageries of Various Spatial Resolution
2. Information on the Margins of a Hard Copy Satellite Image

3. Analysis of Spectral Signatures of Surface Features
4. Visual Interpretation of Features from Satellite Image
5. Digital Satellite Imageries: Data Import & Data Structures
6. Digital Image Interpretation: Band Combination Analysis
7. Digital Image Interpretation: Histogram and Pixel Value Analysis
8. Unsupervised Classification of Satellite Data
9. Supervised Classification of Satellite Data

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Campbell, J. B. and R.H. Wynne, *Introduction to Remote Sensing* (Fifth Edition), Guilford Press, 2012.
2. Jensen, J.R., *Introductory Digital Image Processing: A Remote Sensing Perspective*, Pearson Prentice Hall, 2005.
3. Joseph, G. *Fundamentals of Remote Sensing*, Universities Press, 2005.
4. Lillesand, T.M., R.W. Kiefer and J.W. Chipman, *Remote Sensing and Image Interpretation*, 5th Edition, John Wiley and Sons, 2004.
5. Mather, P.M. and M.G. Koch, *Computer Processing of Remotely-Sensed Images: An Introduction,* John Wiley & Sons, 2011.
6. Prost, Gary L., *Remote Sensing for Geoscientists: Image Analysis and Interpretation*, 3rd Edition, CRC Press, Taylor & Francis, 2013.

Further Readings:

1. Gomarasca, Mario A, *Basics of Geomatics*, Springer: Heidelberg, 2009.
2. Nag, P. and M. Kudrat, *Digital Remote Sensing*, Concept Publishing Company, New Delhi, 1998.
3. Richards, John A. *Remote Sensing Digital Image Analysis: An Introduction,* 5th Edition, Springer-Verlag Berlin Heidelberg, 2013.
4. Gupta, R.P., *Remote Sensing Geology*, Springer, Heidelberg, Germany, 2003.
5. Roy, P.S.; Van Westen, C.J.; Jha, V.K.; Lakhera, R.C. and Champati Ray, P.K., *Natural Disaster and their Mitigation: Remote Sensing and Geographical Information System Perspectives*, IIRS, Dehra Dun, Govt. of India, 2000.
6. Verbyla, David, L., *Satellite Remote Sensing of Natural Resources*, Lewis Publishers, New York, 2005.

PEDAGOGY: Students may be taken to field for verification of identifiable features on satellite imageries; they may be asked to prepare a report using toposheets, satellite imageries and other data sources to trace the temporal and spatial changes. Students may be taken to different institutions to acquaint with different remote sensing equipments, techniques, and products. Remote Sensing professionals may be invited to interact and deliver special lectures to students on remote sensing techniques.

PAPER II: FUNDAMENTALS OF PHOTOGRAMMETRY

Max. Marks	**: 100**
Theory	: 50
Practical	: 30
Internal Assessment	: 20

Objectives: The aim of this paper is to provide a general overview of Photogrammetry, its theory and general working principles; to introduce the basic concepts and significance of Photogrammetry and to provide a practical understanding of principles of measurement from vertical aerial photographs.

(A) THEORY **Marks: 50**
Time: 3 Hours

UNIT I

i. Photogrammetry: Definition and Categories
ii. Historical Background: Early Developments in Aerial Surveying and Mapping
iii. Problems of Aerial Photogrammetry
iv. Application of Photogrammetry

UNIT II

v. Basic Mathematics of Photogrammetry: Geometry of Aerial Photographs
vi. Calculation of Photo Scale,
vii. Relief Displacement: Concept and Calculations
viii. Parallax: Concept and Determination
ix. Calculation of Height of Objects on Vertical Aerial Photograph

UNIT III

x. Stereoscopy: Definition
xi. Principles of vision, Binocular Vision, Stereoscopic Vision Testing,
xii. Pseduoscopic Views
xiii. Types of Stereoscopes and their Operations

UNIT IV

xiv. Photogrammetry in the Space Age
xv. Photogrammetry for Missiles and Satellite Tracking
xvi. Ortho-photos and Contour Extraction

NOTE:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Distinction between Map, Photo and Stereoscopic view.
2. Arrangement of Aerial Photographs with the help of numbers given on the photograph and preparation of Photo Index.

3. Understanding the simple geometry of aerial photographs
4. Interpretation of information given on the margins of photograph and satellite imageries.
5. Locating principal point, conjugate principal point, flight line
6. Stereoscopic vision Test, selection of stereo pair and its orientation and placing under the stereoscope for stereovision.
7. Identification and mapping of features from single photograph.
8. Identification and mapping of features from a stereo pair and their verification in the field.
9. Measurement of scale, relief displacement , parallax , height and slope from vertical aerial photographs
10. Identification of planimetric positions

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings

1. McGlone, J. Chris, *Manual of Photogrammetry (Sixth Edition),* American Society of Photogrammetry and Remote Sensing, 2013.
2. Mikhail, Edward M. et al, *Introduction to Modern Photogrammetry*, John Wiley & Sons Inc. 2001.
3. Moffitt, F. H., *Photogrammetry*, 3rd Edition, Harper & Row, NY, 1980.
4. Paul Wolf et al., *Elements of Photogrammetry with Application in GIS*, Fourth Edition, McGraw-Hill Professional, 2013.
5. Wolf, P. R., *Elements of Photogrammetry*. McGraw-Hill, NY, 1983.
6. Zorn, H. C., *Introductory Course in Photogrammetry*. 6th Ed. ITC, Netherlands, 1980.

Further Readings

1. Burnside, C. D., *Mapping from Aerial Photography*. 2nd Ed, Collins, 1985.
2. Campbell, J.B., *Introduction to Remote Sensing*, 3rd ed., The Guilford Press, 2002.
3. Curran Paul, J., *Principles of Remote Sensing*, UK: ELBS, 1984.
4. Joseph, George, *Fundamentals of Remote Sensing,* Universities Press India, 2007.
5. Lillesand, T. M.; R.W. Kiefer, and J.W. Chipman, *Remote Sensing and Image Interpretation*, 5th Edition, Wiley, 2007.
6. Sabins, Floyd F., *Remote Sensing: Principles and Interpretation,* New York: WH Freeman and Company, 2007.

PEDAGOGY: Students will be demonstrated the difference between qualitative and quantitative understanding of various aspects of Photogrammetry. Demo exercise on measurements on the map and photo will make students aware the importance of Photogrammetry in objective and precise mapping using aerial photographs.

PAPER III: FUNDAMENTALS OF GEOGRAPHIC INFORMATION SYSTEMS (GIS)

Max. Marks	**: 100**
Theory	: 50
Practical	: 30
Internal Assessment	: 20

OBJECTIVES: The main objective of this course is to expose the students to fundamental principles of Geographical Information Systems including basic concepts and definitions, methods and techniques.

(A) THEORY **Marks: 50**
Time: 3 hours

UNIT – I

Concepts and Definitions:

i. Geographic Information Systems (GIS): Definition and Developments
ii. Spatial data base: Points, Lines and Polygons
iii. GIS Components: Hardware and Software

UNIT – II

Functional Elements:

iv. Data Acquisition, Input and Editing
v. Data Manipulation and Topology Creation
vi. Data Analysis and Query

UNIT – III

Data Management and Structure:

vii. Data Base Management System (DBMS): Purpose and structure
viii. Classification: Raster, Vector, Quadtree, Hierarchical, Network and Relational

UNIT - IV

GIS Applications:

ix. GIS Applications
x. GIS: Decision Support System

NOTE:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. GIS Software: Introduction to Arc GIS
2. Georeferencing Maps/Images
3. Digitization of Raster Map: Point, Line and Polygon Features
4. Preparation of Attribute Tables, Editing and Joining Tables
5. Analyzing Attribute Data: Calculating Area, Perimeter, Length

6. Spatial Representation: Mapping Techniques
7. Spatial Representation: Symbolizing and Map Layouts
8. Basic Analysis in GIS: Buffering, Overlay and Query Building

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Harvey, F., *A Primer of GIS: Fundamental Geographic and Cartographic Concepts*, Guilford Press, New York, 2008.
2. Ian Heywood,Sarah Cornelius and Steve Carver, *An Introduction to Geographical Information Systems*, 3rd Edition, Pearson, Dorling Kindersley (India) Pvy. Ltd, 2010.
3. Kang-tsung (Karl) Chang, *Introduction to Geographic Information Systems*, 4th Edition, Tata McGraw-Hill, New Delhi, 2009.
4. Maguire, D.J.; M. Batty and M.F. Goodchild, *GIS, Spatial Analysis and Modeling*, ESRI Press, 2005.
5. Michael F. Goodchild, *Introduction to GIS*, Santa Barbara, California, & Karan K. Kemp (eds.) NGGIA, 1990.
6. Muralikrishna, I.V., *Remote Sensing Applications and Geographic Information Systems - Recent Trends*, Tata McGraw Hill, New Delhi, 1992.
7. Wise, S., *GIS Fundamentals*, 2nd Edition, CRC Press, Taylor & Francis Group, 2013.

Further Readings:

1. Aronoff, S., *Geographic Information Systems: A Management Perspective***,** WDL Publications Ottawa,Canada, 1992.
2. Burrough, P.A. & Macdoneli, R.A., *Principles of Geographic Information Systems,* Oxford University Press, 2000.
3. Demers, M.N., *Fundamentals of Geographic Information Systems*, 2nd Edition, John Wiley and sons, 2000.
4. Jefrey S. & John, E., *Geographic Information Systems - An Introduction*, Prentice Hall, New Jersey, USA, 1990.
5. Laurini, R. & Thomson, D., *Fundamentals of Spatial Information Systems*, Academic Press, London, 1994.
6. Rashid, S.M., *Remote Sensing in Geography*, Manak Publications, Delhi, 1993.
7. Shen, Zhenjiang, *Geospatial Techniques in Urban Planning*, Springer-Verlag Berlin Heidelberg, 2012.

PEDAGOGY: Taking students for fieldwork, inviting professionals for lectures, taking students to institutes/training centers engaged in imparting professional training and providing research consultancy in business, tourism and area/regional planning will be the main component of the teaching methodology.

PAPER-IV: FUNDAMENTALS OF COMPUTERS AND COMPUTER PROGRAMMING

Max. Marks : 100
Theory : 50
Practical : 30
Internal Assessment : 20

OBJECTIVES: To develop basic skills and understanding of computer operations; to develop skills of Word Processing for written communication of results and documentation of work performed; to impart skills for data management, charting, and visualization; to expose students to the effective techniques of using search engines & the services of Internet and to impart programming skill for developing programs in 'C' language.

(A) THEORY **Marks: 50**
Time: 3 Hours

UNIT-I

Computer Concepts:
i. Introduction to Computers
ii. Computer System, Hardware, Software, Input and Output Devices
iii. Data Representation
iv. Fundamental Concepts of Operating Systems.
OS and Application Software:
v. Introduction to Windows Operating System.
vi. Introduction to MS-Word and MS-Excel Software.

UNIT-II

Computer Graphics:
vii. Fundamentals of Computer Graphics, Image Formats
Computer Networks:
viii. Introduction to computer networks
ix. Overview of Internet and its services,
x. Search Engines, Techniques to search effectively.
Web Application Development:
xi. Introduction to web application development using HTML.

UNIT-III

Data concepts:
xii. Key Database Concepts
xiii. Database Management Systems, Type of DBMS
xiv. Structured Query Language (SQL)
xv. Data Visualization, Data Mining.
Information Systems
xvi. Introduction to Information Systems
xvii. Types of Information Systems.

UNIT-IV

Programming Concepts:
xviii. Concept of a program, data types, keywords, structure of a program.
Writing 'C' programs:
xix. Statements, Types of Statements
xx. Library Functions and User Defined Functions
xxi. Arrays
xxii. File Handling.

NOTE:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Hands on experience on Windows operating system.
2. Hands on experience on MS-Word
3. Hands on experience on MS-Excel
4. Creating web pages using HTML,
5. Techniques to use search engines effectively.
6. Writing queries using SQL.
7. Writing C programs.

Note: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the department, and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings

1. Miller, M., *Easy Computer Basics, Windows 8.1 Edition*, Pearson Education, Inc. 2014.
2. Basandra, S. K., *Computers Today*, Galgotia Publications Pvt. Ltd., 2010.
3. Salaria, R.S. *Computer Concepts and Programming in C*, Salaria Publishing House, 2013.
4. Lambert, J. and J. Cox, *Microsoft Word 2013: Step by Step*, Eastern Economy Edition, 2013.
5. Frye, C., *Microsoft Excel 2013: Step by Step*, Microsoft Press, 2013.
6. Online Resources: www.w3schools.com for HTML and SQL. (*Live and latest).*

Further Readings

1. Freedman, J., *Microsoft Word 2013: Plain & Simple*, O'Reilly Media Inc., 2013.
2. Norton, P., *Introduction to Computers*, 6th Edition, McGraw-Hill Education, 2004.
3. Norton, P., *Computing Fundamentals*, Student Edition, McGraw-Hill, 2004.

PEDAGOGY: Bottom up approach of explaining concepts; programming exercises to simulate interest to practice programming applications; Audio visual presentations; Test/Assignment as per the latest in industry/real-life. Students will be encouraged to ask questions and participate in the classroom discussion.

SEMESTER-II

PAPER V: CARTOGRAPHY

Max. Marks : 100
Theory : 50
Practical : 30
Internal Assessment : 20

OBJECTIVES: The course is designed to emphasize the relevance and scope of cartography in GIS. It shall highlight the value of this vital field in the current and highly versatile field of Remote Sensing and GIS.

(A) THEORY **Marks: 50**
Time: 3 Hours

UNIT – I

Basic Concepts of Cartography

i. Cartography: Definition, Scope and Significance
ii. Historical Perspective of Cartographic Development
iii. Traditional and Digital Cartography
iv. Cartography and GIS

UNIT – II

Data Types and Map Basics

v. Levels of Measurements
vi. Data: Types (Spatial and Attribute data), Sources and Analysis
vii. Map Types and Classifications; Topographic Maps
viii. Map Projections, Datum and Coordinate System

UNIT – III

Map Designs and Production

ix. Principles of Symbolization
x. Basics of Colour Schemes and Representation
xi. Map Lettering and Legend Presentation
xii. Map Layout and Product Generation

UNIT – IV

Interpretation of Topographical Maps

xiii. Relief Representation: Contours, Hachures, Hill Shading
xiv. Relief (Absolute and Relative) and Profiles (Longitudinal and Transverse)
xv. Slope Analysis (Wentworth and Robinson)
xvi. Reading Topographical Maps

NOTE:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Classification of Topographical Sheets
2. Slope Analysis (Wentworth and Robinson)

3. Generation of Hill Shading Map
4. Profiles: Longitudinal and Transverse Profiles
5. Interpretation of Physical and Cultural features from Topographical Sheet
6. Thematic Maps: Dot Map, Choropleth Map, Proportionate Circles, Isopleth Map

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings

1. Cromley, R.G., *Digital Cartography*, Prentice-Hall of India, New Delhi, 1992.
2. Crampton, W.C., *Mapping: A Critical Introduction to Cartography and GIS,* John Willy & Sons, New York, 2010.
3. Dent, B.D., *Cartography-Thematic Map Design*, WBC McGraw–Hill, Boston, 1999.
4. John Krygier and Denis Wood, *Making Maps: A visual guide to map design for GIS,* Guilford Publications, New York, 2005.
5. Keates, J.S., *Cartographic Design and Production*, Longman, London, 1998.
6. Misra, R.P. and Ramesh, A., *Fundamental of Cartography*, Concept Publishing Company, New Delhi, 1989.
7. Monkhouse, F.J., *Maps and Diagrams*, Methuen and Co., London, 1994.
8. Nag, P., *Thematic Cartography and Remote Sensing*, Concept Publishing Co., New Delhi ,1992.
9. Nelson Petrie*, Analysis and Interpretation of Topographical Maps (Rev),* Orient Blackswan Pvt. Ltd. 1997.
10. Rampal, K.K., *Mapping and Compilation*, Concept Publishing Co., New Delhi, 1993.
11. Raisz, Erwin, *Principles of Cartography*, McGraw Hill, New York, 1962.
12. Robinson, A.H. and others, *Elements of Cartography*, John Willy & Sons, New York, 6th edition, 2009.
13. Singh, R.L. and Singh R.P.B., *Elements of Practical Geography,* Kalyani Publishers , New Delhi, Reprint 2002.

Further Readings:

1. Birch, T., *Maps- Topographical and Statistical*, Clarendon Press, Oxford, 1949.
2. Brown, L.A., *The Story of Maps,* Cressit Press, London, 1951.
3. Harley J.B. and David Woodward, *The History of Cartography* Volume 2, Book 2, The University of Chicago Press, 1995.
4. Raisz, Erwin, *Principles of Cartography*, McGraw Hill, New York, 1962.
5. Ramamurthy, K., *Maps Interpretation: India Landscapes Through Survey of India*, Topographic Maps, R.K. Mutt Road, Madras, 1982.
6. Tyner, J.A., *Principles of Map Design*, Guilford Publications, New York, 2010.
7. Wood Clifford H. and Keller C., *Cartographic Design- Theoretical and Practical Perspectives*, John Wiley & Sons, 1996.

PEDAGOGY: Students will be encouraged to ask questions and participate in classroom discussions. Assignments, term papers and group discussions will be the additional methods to make teaching more focused and interesting.

PAPER-VI: SPATIAL ANALYSIS

Max. Marks : 100
Theory : 50
Practical : 30
Internal Assessment : 20

OBJECTIVES: The course is designed to help the students to understand the techniques, which assist in providing focused and precise analysis of emerged pattern of geographical phenomena.

(A) THEORY **Marks: 50**
Time: 3 Hours

UNIT- I

Fundamentals of Spatial analysis

i. Significance of Spatial analysis
ii. Nature of Spatial Data
iii. Using GIS for Spatial Analysis.

UNIT- II

Point Pattern Analysis

iv. Point Features Over Space: Meaning and Nature
v. Measures of Central tendency: Mean Centre, Weighted Mean Centre, Median Centre
vi. Measures of Dispersion: Standard distance, Standard Deviation Ellipse, Nearest Neighbour Analysis
vii. Spatial Autocorrelation

UNIT –III

Line Pattern Analysis

viii. Linear Features Over Space: Meaning and Nature
ix. Network Analysis: Connectivity or Topology

Polygon Pattern Analysis

x. The Meaning of Spatial Relationships and Spatial Dependency
xi. Spatial Autocorrelation

UNIT-IV

Vector Analysis

xii. Buffering: Point, Line, Area buffers
xiii. Overlay: Union, Intersect, Symmetrical Difference, Identity, Point-in-Polygon, Line-in-Polygon, Polygon-in-Polygon

Surface Analysis

xiv. Digital Elevation Model (DEM) and Digital Terrain Model (DTM)

NOTE:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Familiarization with Spatial Analysis Software.
2. Measuring Mean Centre, Weighted Mean Centre and Median Centre
3. Measuring Standard Distance and Standard Distance Ellipse
4. Nearest Neighbour Analysis
5. Spatial Autocorrelation of Point Features
6. Network Analysis
7. Vector Analysis: Buffering, Overlay and Query
8. Surface Analysis: Digital Elevation Model (DEM), Digital Terrain Model (DTM)

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Chang, Kang-tsung, *Introduction to Geographic Information Systems,* Sixth Edition, Tata McGraw-Hill, 2011.
2. Fotheringham Stewart A and Rogerson A. Peter, *The SAGE Handbook of Spatial Analysis*, SAGE Publications, London, 2009.
3. Laurini, R. & Thomson, D., *Fundamentals of Spatial Information Systems*, Academic Press, 1994.
4. Sanders, L., *Models in Spatial Analysis*, John Wiley & Sons, 2013.
5. Wong, David W.S. & Jay Lee, *Statistical Analysis of Geographic Information with ArcView GIS and ArcGIS*, John Wiley and Sons, 2005.

Further Readings:

1. Burrough, P.A. & Macdoneli, R.A., *Principles of Geographic Information Systems,* Oxford University Press, 2000.
2. Demers, M.N., *Fundamentals of Geographic Information Systems*, 2nd Edition, John Wiley and sons, 2000.
3. Maguire, D.J.; M. Batty and M.F. Goodchild, *GIS, Spatial Analysis and Modeling*, ESRI Press, 2005.
4. Murayama, Y. and R. Thapa, *Spatial Analysis and Modeling in Geographical Transformation Process: GIS Based Applications*, Springer, 2011.

PEDAGOGY: Students may be asked to prepare a report on spatial analysis applications using external and self generated databases. Students will be encouraged to ask questions and participate in classroom discussions.

PAPER VII: IMAGE PROCESSING

Max. Marks	**: 100**
Theory	: 50
Practical	: 30
Internal Assessment	: 20

OBJECTIVES:

The course is designed to introduce the student to analytical tools and methods which are currently used in digital image processing as applied to image information for human viewing. Then apply these tools in the laboratory in image restoration, enhancement and compression.

(A) THEORY **Marks: 50**

Time: 3 Hours

UNIT – I

i. Digital Image Pre-processing: Sources of Errors, Types of Error- Radiometric and Geometric

ii. Image Rectification: Geometric Correction, Radiometric Correction, and Noise Removal.

UNIT - II

iii. Image Enhancement Techniques: Contrast Enhancement; Linear Contrast Stretch, Histogram Equalisation, Gaussian Stretch, Density Slicing, Pseudocolour Transform.

UNIT - III

iv. Image Transformation: Band Ratioing, Principal Component Analysis, Hue, Saturation and Intensity Transformation.

v. Spatial Filtering: Low Frequency and High Frequency, Edge Enhancement.

UNIT - IV

vi. Digital Image Classification: Supervised; Unsupervised; Accuracy Assessment.

NOTE:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Familiarization with Image Processing Software
2. Georeferencing/Image Registration
3. Layer Stacking
4. Image Mosaic Preparation & Image Subset
5. Image Enhancement & Transformation Techniques
6. Filtering Techniques
7. Supervised Classification
8. Unsupervised Classification
9. Accuracy Assessment

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Burger W., & Burge, M.J., *Principles of Digital Image Processing,* Springer-Verlag London Limited, 2009.
2. Campbell, J. B. and R.H. Wynne, *Introduction to Remote Sensing* (Fifth Edition), Guilford Press, 2012.
3. Chen, C.H., *Image Processing for Remote Sensing,* CRC Press, 2008.
4. Gonzalez, R.C. & Woods, R.E., *Digital Image Processing*, Second edition, Prentice-Hall, Inc., 2002.
5. Jensen, J.R., *Introductory Digital Image Processing: A Remote Sensing Perspective*, Prentice Hall, New Jersey, 1986.
6. Liu, J.G. & Mason, P.J., *Essential Image Processing and GIS for Remote Sensing,* John Wiley & Sons, 2009.
7. Mather, P.M. & Magaly, K., *Computer Processing of Remotely-Sensed Images: An Introduction,* Fourth Edition, John Wiley & Sons, 2011.
8. Richards, John A. *Remote Sensing Digital Image Analysis: An Introduction,* 5th Edition, Springer-Verlag Berlin Heidelberg, 2013.

Further Readings:

1. Gomarasca, Mario A, *Basics of Geomatics*, Springer: Heidelberg, 2009.
2. Gupta, R.P., *Remote Sensing Geology*, Springer, Heidelberg, Germany, 2003.
3. Jain, A.K., *Fundamentals of Digital Image Processing,* Prentice-Hall, Inc., 2002.
4. Joseph, G. *Fundamentals of Remote Sensing*, Universities Press, 2005.
5. Lillesand, T.M., Kiefer, R.W. & Chipman, J. W., *Remote Sensing and Image Interpretation*, Fourth Edition, John Wiley and Sons, 2004.
6. Pratt, W.K., *Digital Image Processing*, Third Edition, John Wiley & Sons, 2001.
7. Prithvish Nag and M. Kudrat, Digital *Remote Sensing*, Concept Publishing Company, New Delhi, 1998.
8. Prost, Gary L., *Remote Sensing for Geoscientists: Image Analysis and Interpretation*, 3rd Edition, CRC Press, Taylor & Francis, 2013.
9. Verbyla, David, L., *Satellite Remote Sensing of Natural Resources*, Lewis Publishers, New York, 2005.

PEDAGOGY: Students may be asked to prepare a report on image processing techniques using satellite images. Students will be encouraged to ask questions and participate in classroom discussions.

PAPER VIII: FUNDAMENTALS OF GLOBAL POSITIONING SYSTEM (GPS)

Max. Marks	**: 100**
Theory	: 50
Practical	: 30
Internal Assessment	: 20

Objectives: The objectives of this paper are to give a comprehensive introduction to GPS and DGPS technology, system concepts, design, operation, implementation and applications, to provide detailed information on the GPS signal, the processing by the receiver, and the techniques by which GPS obtains position, velocity and time.

(A) THEORY **Marks: 50**
Time: 3 Hours

UNIT I

i. Concept and Characteristics of Global Navigational Satellite System (GNSS)
ii. Historical Background, GPS Segments, Features of GPS Satellites, Principles of Operation, Fundamentals on Satellite Geodesy, Referential Systems related to GNSS, Survey and Characteristics of Observed Values.

UNIT II

iii. Surveying with GPS: Absolute, Relative and Differential Positioning, Kinematic GPS.
iv. Performing GNSS Measuring and Data Processing, Overview on Geodetic Data and Data Transformation

UNIT III

v. Survey on GNSS Application: Basic Principles in Working with DGNSS-Mathematical Models, Coordinate Systems in Function, Practical Problems-Navigation using GNSS
vi. Methods for Determining and Techniques for Searching Ambiguities both for Phase Data and the Combination of Code and Phase Data GNSS Application in: Geodesy, Geodynamics, Tectonic Plate Movement with Time Alternating Coordinates, Navigation, Space.

UNIT IV

vii. Diverse Systems for Global Positioning: TRANST, DORIS, GLONASS, GPS, GALILEO, GAGAN.
viii. Location-based Services: Practical Application.

NOTE:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **30 Marks**

1. Components and Operation of GPS Instrument
2. Understanding the Different Pages of GPS
3. Collecting Readings using Primary Pages
4. Surveying a Small Area
5. Mapping of the Surveyed Area
6. Tabulation of Information of the Surveyed Area
7. Analysis of the Map and Data Collected
8. Application Exercises

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Anderle, R., *The Global Positioning System*, Royal Society of London, UK, 1988.
2. Blewitt, G., *Advances in GPS Technology for Geodynamics Investigations: 1978-1992*, Crustal Dynamics Project AGU Monograph, NASA. 1993.
3. Colombo O. L. & M. M. Watkins, *Satellite Positioning*, in US National report to IUGC on Geodesy, 1991.
4. C. Jones, *Geographical Information Systems and Computer Cartography*, Pearson Education Inc, 1997.
5. C. Rizos, *Introduction to GPS*, University of New South Wales, 1999.
6. El-Rabbani, Ahmed, *Introduction to GPS: The Global Positioning System*, Artech House, 2002.
7. Kaplan, Elliott D*., Understanding GPS: Principles and Applications*, Artech House, 2005.
8. Letham, Lawrence, *GPS Made Easy: Using Global Positioning Systems in the Outdoors*, Rocky Mountain Books, 2008.

Further Readings:

1. Goad, C. C., *Proceedings of First International Symposium on GPS*, USA, 1985.
2. Fell, P. J., *Geodetic Positioning Using GPS*, The Ohio State University Report no. 299, Columbus, Ohio, USA, 1980.
3. King, R. W. & Others, *Surveying GPS*, University of New South Wales, Australia, 1985.

PEDAGOGY: Students will be provided demo exercises to increase their awareness about wide ranging applications of GPS.

SEMESTER III

PAPER IX: REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS IN GEOSCIENCES

Max. Marks	**: 100**
Theory	: 50
Practical	: 30
Internal Assessment	: 20

OBJECTIVES: To train the students to detect, identify and analyze the various facets of geomorphic and geological features; analyze various hydrological and geo-environmental parameters and manage disasters with the help of remote sensing and GIS.

(A) THEORY **Marks: 50**
Time: 3 Hours

UNIT - I

Remote Sensing and GIS in Geological and Geomorphological Studies

i. Terrain Elements; Spectral Signatures of Rocks
ii. Lithological; Structural and Geomorphological mapping
iii. Thermal and Microwave Remote Sensing for Geological Mapping
iv. RS and GIS for Mineral and Oil Exploration

UNIT - II

Remote Sensing and GIS for Hydrological and Geo-environmental Studies

v. Surface Water and Ground Water Management
vi. Morphometric Analysis of a Drainage Basin
vii. Soil and Land Degradation
viii. Land Use/Land Cover Analysis

UNIT – III

Remote Sensing and GIS in Disaster Management

ix. Hazard, Vulnerability, Risk, Disaster
x. Types of Hazards/Disasters
xi. Disaster Management
xii. Role of Geoinformatics in Disaster Management

UNIT – IV

Remote Sensing and GIS in Hazard and Risk Analysis

xiii. Remote Sensing Data for Disaster Studies
xiv. Hazard Mapping and Analysis
xv. Risk and Vulnerability Assessment

NOTE:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Mapping and Analysis of Absolute and Relative Relief
2. Measurement of Slopes: Degree, Aspect and Curvature
3. Generation of Spatial Profile and Surface Profile
4. Generation of 3-Dimensional Model of Terrain
5. Morphometric Analysis of Drainage Basins
6. Land Use/Land Cover Analysis
7. Geomorphological Mapping
8. Lineaments/Fault Mapping and Analysis
9. Landslide Hazard Zonation
10. Flood Hazard and Risk Analysis

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Campbell, J. B. and R.H. Wynne, *Introduction to Remote Sensing* (Fifth Edition), Guilford Press, 2012.
2. Gupta, R.P., *Remote Sensing Geology*, Springer, Heidelberg, Germany, 2003.
3. Hyndman, D. and D. Hyndman, *Natural Hazards and Disasters*, 2nd edition. USA, Belmont: Brooks/Cole, 2009
4. Roy, P.S.; Van Westen, C.J.; Jha, V.K.; Lakhera, R.C. and Champati Ray, P.K., *Natural Disaster and their Mitigation: Remote Sensing and Geographical Information System Perspectives*, IIRS, Dehra Dun, Govt. of India, 2000.
5. Verbyla, David, L., *Satellite Remote Sensing of Natural Resources*, Lewis Publishers, New York, 2005

Further Readings:

1. Agarwal, C.S. and P.K. Garg, *Text Book on Remote Sensing in Natural Resources Monitoring and Management*, Wheeler Publishing Co, New Delhi, 2000.
2. Greedry, Alan, F., *Application of Remote Sensing with Special Reference to Geosciences*, Gregory Geo-science, 1974.
3. Papova, T.A. and Pandey, S.N., *Principles and Applications of Photogeology*, Wiley Eastern Publishers, 1987
4. Smith, Willam, L., *Remote Sensing Applications for Mineral Exploration*, Dawden Hutchintons and Ross Inc., 1977.
5. Verstappen, H., *Remote Sensing in Geomorphology*, Elsevier Scientific Publications, Netherlands, 1977.

PEDAGOGY: Students may be taken to institutions such as IIRS, NRSC and State Remote Sensing Centers to acquaint them with equipments, techniques and their products. Students may be asked to prepare a report on landform using topographical sheets, aerial photographs and satellite images. Students may acquaint with the satellite imageries of various kinds of environmental problems.

PAPER X: REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS IN URBAN AND REGIONAL PLANNING

Max. Marks	**: 100**
Theory	: 50
Practical	: 30
Internal Assessment	: 20

OBJECTIVES: To demonstrate the application of Remote Sensing and GIS technologies in urban and regional planning

(A) THEORY **Marks : 50**
Time: 3 Hours

UNIT – I

Basic Concepts

i. Planning: Definition and Types
ii. Concept of Urban and Urban Planning
iii. Concept of Region and Regional Planning

UNIT – II

Urban Planning

iv. Role of Remote Sensing and GIS in Urban Planning
v. Creation of Data Base for Urban Planning
vi. Urban Land Use Maps; Location of Facilities and Amenities

UNIT – III

Regional Planning

vii. Role of Remote Sensing and GIS in Regional Planning
viii. Creation of Data Base for Regional Planning
ix. Land Use/Land Cover Analysis; Land Use Suitability Evaluation

UNIT – IV

Urban and Regional Management

x. Urban Change Detection and Urban Growth Monitoring
xi. Slum Identification and Improvement; Urban Policing.
xii. Application of GIS for Perspective Urban and Regional Planning

NOTE:

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

(B) PRACTICAL **Marks: 30**

1. Preparation of Base Map for Regional and Urban Planning.
2. Location of Basic Amenities and Facilities
3. Land Use and Land Cover Analysis.
4. Urban Growth Monitoring and Change Detection.
5. Slum Identification and Mapping.
6. Urban Crime Analysis

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Bhatta, B., *Analysis of Urban Growth and Sprawl from Remote Sensing Data*, Springer-Verlag Berlin Heidelberg, 2010.
2. Campbell, J. B. and R.H. Wynne, *Introduction to Remote Sensing* (Fifth Edition), Guilford Press, 2012.
3. Chang, Kang-tsung, *Introduction to Geographic Information Systems,* Sixth Edition, Tata McGraw-Hill, 2011.
4. Muralikrishna, I.V., *Remote Sensing Applications and Geographic Information Systems - Recent Trends*, Tata McGraw Hill, New Delhi, 1992.
5. Ian Heywood,Sarah Cornelius and Steve Carver, *An Introduction to Geographical Information Systems*, Prentice Hall College Div, New Jersey, USA, 2000.
6. Kang-tsung (Karl) Chang, *Introduction to Geographic Information Systems*, 4th, Edition, Tata McGraw-Hill, New Delhi, 2009.
7. Shen, Zhenjiang, *Geospatial Techniques in Urban Planning*, Springer-Verlag Berlin Heidelberg, 2012.
8. Wong, David W.S. & Jay Lee, *Statistical Analysis of Geographic Information with ArcView GIS and ArcGIS*, John Wiley and Sons, 2005.

Further Readings:

1. Barret, E.C. & Curtis, E.F., *'The Built Environment' in Remote Sensing of Environment*, Second edition, pp. 293-306, 1982.
2. Brimicomne, A. GIS, *Environmental Modeling and Engineering*, 2nd Edition, CRC Press: Taylor & Francis Group, 2010.
3. Burrough, P.A. & Macdoneli, R.A., *Principles of Geographic Information Systems,* Oxford University Press, 2000.
4. Demers, M.N., *Fundamentals of Geographic Information Systems*, 2nd Edition, John Wiley and sons, 2000.
5. Maguire, D.J.; M. Batty and M.F. Goodchild, *GIS, Spatial Analysis and Modeling*, ESRI Press, 2005
6. Sundaram, K.V., *Urban and Regional Planning in India*, Concept, New Delhi, 1977.
7. Taylor, John, L., *Urban Planning Practice in Developing Countries*, Williams, David C Pergamon Press, 1981.
8. Verbyla, David, L., *Satellite Remote Sensing of Natural Resources*, Lewis Publishers, New York, 2005.

PEDAGOGY: Taking students for fieldwork, inviting professionals for lectures, taking students to institutes/training centers engaged in imparting professional training and providing research consultancy in urban and regional planning should be the main focus.

PAPER XI: RESEARCH METHODOLOGY

Max. Marks : 100
Theory : 80
Internal Assessment : 20

OBJECTIVES: The aim of this paper is to train the students about the various aspects of research writing so that they are in a position to write research reports independently.

(A) THEORY **Marks : 80**
Time: 3 Hours

UNIT-I

Basic Concepts

i. Meaning, Objectives and Significance of Research
ii. Types of Research and Research Approaches.
iii. The Research Process

UNIT-II

Data in Research

iv. Types and Sources of Data
v. Data collection: Methods and Techniques
vi. Data Processing and Analysis

UNIT-III

Sampling:

vii. Sampling: Concept, Need and Significance
viii. Probability and Non-Probability Sampling
ix. Sampling Techniques

UNIT-IV

Research Proposal and Report Writing

x. Formulation of Research Proposal
xi. Types and Elements of Report;
xii. Steps involved in Report Writing ; Presentation of Findings

NOTE:

1. A Compulsory Question at serial number I containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 15 marks each (total 60 marks). These will be in addition to the Compulsory Question at serial number I.

(B) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

LIST OF READINGS

Essential Readings:

1. Creswell, John W., *Research Design; Qualitative, Quantitative and Mixed Methods Approach*, SAGE Publications, Los Angeles, 2008.
2. Flick, U. *An Introduction to Qualitative Research*, 5th Edition, SAGE, 2014.
3. Kothari, C. R., *Research Methodology, Methods & Techniques,* New Age International Publisher, N. Delhi, 2008.

4. Kumar Ranjit, *Research Methodology: A step-by-step Guide for Beginners*, SAGE Publications, Ltd. London (Third Edition), 2010.
5. Montello, D. and P. Sutton, *An Introduction to Scientific Research Methods in Geography and Environmental Studies*, SAGE, 2012

Further Readings:

1. Bennet P. Lientz and Kathryn P.**,** *Project Management for the 21st Century*, Academic Press, California, 1995.
2. Earickson, R., and Harlin, J., *Geographic Measurement & Quantitative Analysis*, Macmillan, New York, 1994.
3. Misra, H. N and V. P. Singh, *Research Methodology in Geography, Social, Spatial and Policy Dimensions*, Rawat Publications, N. Delhi, 1998.
4. W. E. Huxold & A. G.Lerinsons Aronoft. S., *Managing Geographic Information Projects*, 1989.

PEDAGOGY: Assignments pertaining to a relevant problem shall be given to students to apply the various aspects of research.

PAPER-XII: INTERNSHIP AND FIELD REPORT

Max. Marks : 100

Duration of Internship: 4 Weeks
Timing of Internship: During Summer Vacation following Semester -II Examination

Distribution of Marks:

a)	Evaluation of the Internship and Field Report	: 50 marks
b)	Presentation of Data Analysis and Main Findings	: 30 marks
c)	Viva-voce	: 20 marks

Evaluation:

The Internship and Field Report shall be prepared by the student on the basis of work done or experiences gained through visit(s) to Govt. Institutes/ Companies, Consultancies, NGOs, Govt. Agency and Other Agencies, Institutes and Centers in India.

The student will submit three copies of his/her work ten days before the commencement of the theory examination of the said semester. Three examiners shall comprise the Board of Examiners consisting of Supervisor, Chairperson of the Department and one more Faculty Member teaching the Course. They shall evaluate the Internship and Field Report independently and the student shall be awarded the mean score of the three evaluations.

The evaluation process shall have three components: (i) Evaluation of the Internship and Field Report, (ii) Presentation of Data Analysis and Main Findings, and (iii) Viva- voce before the Board of Examiners.

Arrangement of Internship:

The Board of Control of Department will plan the task of Internship and coordinate with Govt. Institutes/ Companies, Consultancies, NGOs, Govt. Agency and Other Agencies, Institutes and Centers in India.

Candidates desirous of identifying an agency on their own may submit to the Department a letter stating their intent and a letter of acceptance from the said agency. Those who wish the Department to arrange Internship may express this through a letter to the Chairperson.

The Board of Control will consider these applications and make arrangements in the best interest of the students. All those agencies where Internship will be allowed and undertaken will certify the Attendance, Joining and Relieving of the students/candidates.

SEMESTER IV

PAPER- XIII: PROJECT REPORT

Marks: 400

1. Formulation of Project Proposal : 50 marks 2. Mid Term appraisal : 50 marks 3. Pre-submission Presentation : 50 marks	**Examination Board:** Chairperson Supervisor One Faculty teaching the course
4. Evaluation : 150 marks 5. Presentation : 50 marks 6. Viva-voce : 50 marks	**Examination Board:** Chairperson Supervisor External Examiner
Three copies of the Report shall be submitted only on the approval of the Supervisor; the signatures of the supervisor are mandatory for submission of Project Report.	
A Report Summary shall be submitted with the Project Report in hard and soft copy.	
Project Report to be submitted 3 weeks prior to commencement of University Semester Examination.	

NOTE:

1. **Assigning the Project Title:** Candidates shall submit in the department office their project titles after due consultation with their supervisors within 10 days of commencement of academic session
2. **Formulation of Project Proposal:** Candidates are required to give a presentation on project proposal including the research objectives, methodology and the chapter scheme three weeks after the assigning of the project title. The Board of Examiners consisting of the Chairperson of the Department, Supervisor and one Faculty Member teaching the course shall evaluate the student independently and student shall be awarded the mean score of the three evaluations.
3. **Mid-term Appraisal:** Mid-term appraisal of the project report of each student shall be done by the Board of Examiners within a month of formulation of project proposal. The Board of Examiners consisting of the Chairperson of the Department, Supervisor and one Faculty Member teaching the course shall evaluate the student independently and student shall be awarded the mean score of the three evaluations.
4. **Pre-submission Presentation:** Pre-submission Presentation shall be done by the Board of Examiners within a fortnight of the mid-term appraisal. The Board of Examiners consisting of the Chairperson of the Department, Supervisor and one Faculty Member teaching the course shall evaluate the student independently and student shall be awarded the mean score of the three evaluations.
5. **Evaluation Process:** The evaluation process shall have three-components: (i) evaluation of report, (ii) presentation of data analysis and main findings and (iii) viva voce examination. Evaluation, Presentation and Viva-voce shall be done by the Board of Examiners consisting of the Chairperson of the Department, Supervisor and one External Examiner. They shall evaluate the field report, submitted by the student, independently and student shall be awarded the mean score of the three evaluations.

The project report shall incorporate the findings arrived at on the basis of the data/information collected from the field and processed, analyzed and mapped in the GIS lab.

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